

CrimeFIR

FIR Intelligence & Crime Pattern Detector

Auto-drafts the crime classification of every FIR, links FIRs across police stations and districts by hard evidence, flags repeat-offender clusters and writes station crime briefs, queried in plain English through IBM Bob.

Team Code & Chaos · Track 4: AI & Predictive · Problem Statement 10 · IBM x NFSU Bob Hackathon

IBM Bob · IBM Granite on watsonx.ai · IBM Granite Embedding · Laya · NetworkX

THE PROBLEM

Crores of FIRs, no intelligence layer

- UP Police CCTNS alone holds 3+ crore digitised FIRs, with no NLP layer. Pattern analysis is manual.
- Jamtara-style gangs reuse the same phones and mule accounts against victims in many districts.
- Each victim files at their own station; each FIR writes 98765 43210 / +91-9876543210 / 09876543210 differently.
- Crime type and method (NCRB I.I.F.-II) are filled in by hand, later; the FIR itself has no crime-type field.
- Result: inter-district links are never surfaced, and money in mule accounts is gone before anyone connects cases.

Investigating officer

cannot see their case is the 6th of a ring

Station House Officer

no timely picture of rising crime / active gangs

Victims

₹2,140 cr lost to digital-arrest scams alone (2024)

OUR SOLUTION

Evidence first, AI where it helps, a human decides

1 Rules: hard evidence

phones (all Indian formats), bank accounts, UPI IDs, IMEIs, vehicles, handles, with role: offender / property / complainant

2 Laya: System 1 (local GPU)

crime Major/Minor Head, 24 MO flags, victim gender in one pass, with calibrated confidence
accept if confidence ≥ 0.40

3 IBM Granite: System 2

watsonx.ai granite-4-h-small, only for ~12% low-confidence FIRs; strict JSON, validated, names must appear in the FIR

PS10 asks

Categorise each FIR by crime type

Extract accused, location, MO, victim profile

Detect repeat-offender signatures

Station trend summary + flagged offender list

CrimeFIR delivers

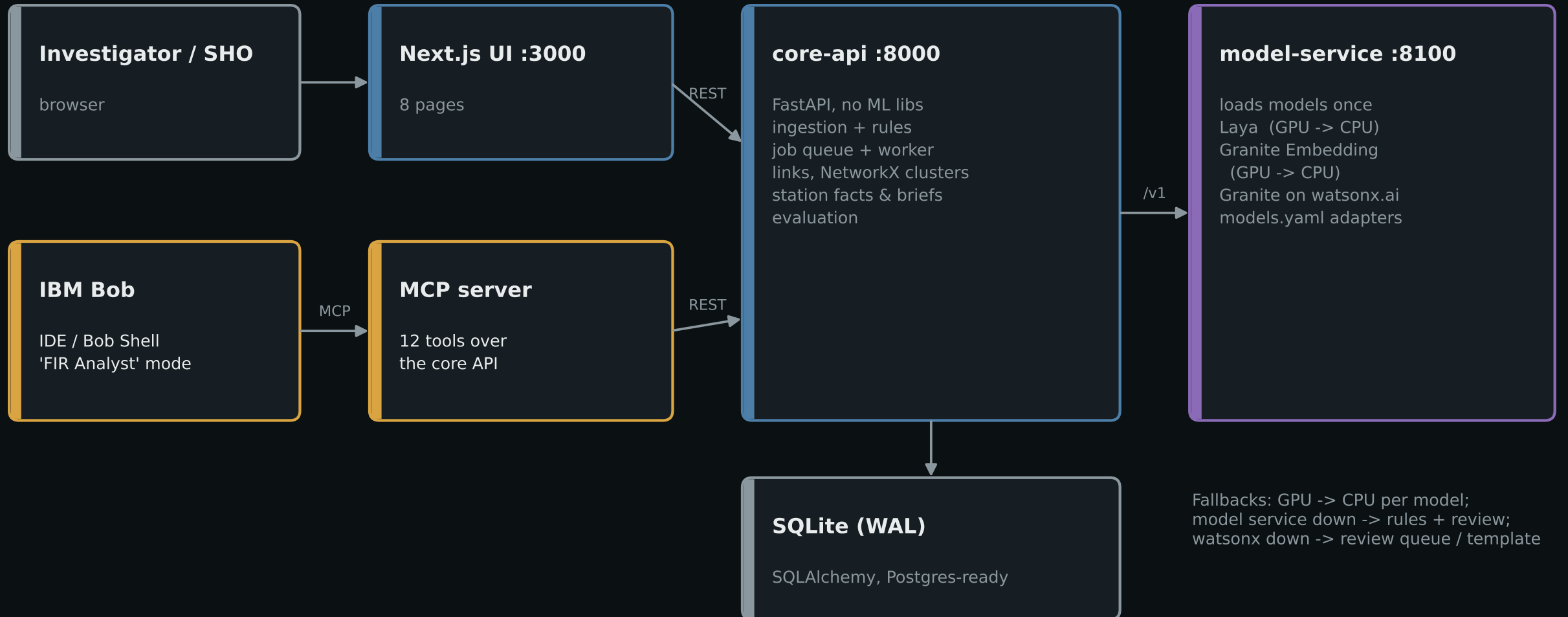
Auto-drafted NCRB I.I.F.-II (4 major, 17 minor heads)

Accused + aliases, place, 24 MO flags, victim, loss

Evidence links + NetworkX clusters with risk score

Number-verified Granite brief + ranked clusters

Architecture



What happens to one FIR



Case file: auto-drafted I.I.F.-II with evidence

CrimeFIR

All investigations & crime pattern detector

Dashboard

Report FIRs

Case Files

Report Offenders

Investigation Graphs

Station Briefs

Model Quality

Satellite - Ahmednagar City

AHM-SAT-2026-0221

Digital arrest / official impersonation reported at Satellite PS (Ahmednagar City) on 24 Sep 2026. Loss about Rs 3,546,500. Method: impersonated police or agency, threat of arrest or case, money to make account.

Report of Nagaiah Jadhav K-001

Graph around this FIR

AUTO-DRAFTED CRIME DETAILS (NOI LX-16)

CRIME OVERVIEW

Digital arrest / official impersonation

Reported by Case: NPS confidential

Alternative digital arrest / official impersonation RMs: OFFENSE & bank impersonation Fraud RN - Criminal Intimidation RN

Method (NOI):
The offender pretended to be police, CB, customs, officer or another government officer (7%), the victim was threatened with arrest, a police case or legal action (75%), money was transferred to a bank account or UPI ID controlled by the offenders (95%).

Victim profile:
Age 37 (36-40)
Gender: male
Occupation: textile trader
Loss: Rs 3,546,500

Account:
Inspector Vikram Rathore (identity claimed by offender)

Related FIRs

Evidence links share a hand identifier pattern links only bank details.

SAR-804-2026-0241

Digital arrest / official impersonation - Adgaon Bawal City - 21 Aug 2026

Same bank account #0230517360305 (written as "40230517360305" and "40230517360305")

Offender claimed the same identity in both "Vikram Rathore" (supporting only)

Investigation lead only. Requires human verification.

VAD-SAY-2026-0295

Digital arrest / official impersonation - Sayajpuri (Mumbai City) - 22 Aug 2026

Same bank account #0230517360305 (written as "40230517360305" and "40230517360305")

Offender claimed the same identity in both "Vikram Rathore" (supporting only)

Investigation lead only. Requires human verification.

AMP-SAT-2026-0236

Digital arrest / official impersonation - Satellite (Ahmednagar City) - 26 Jul 2026

Same bank account #0230517360305 (written as "40230517360305" and "40230517360305")

Offender claimed the same identity in both "Vikram Rathore" (supporting only)

Investigation lead only. Requires human verification.

AMP-CYS-2026-0258

Digital arrest / official impersonation - Cyber Crime Police Station (Ahmednagar City) - 28 Jul 2026

Same crime type: Digital arrest / official impersonation

Reportation via RMs: similar identity impersonation

Same multiple impersonated police or agency, money to make account, threat of arrest or case

30 days apart

No shared evidence pattern similarity only

Investigation lead only. Requires human verification.

First information contents

Highlighted evidence extracted by rules

The complainant, a textile trader aged 57 years, states that he got a WhatsApp video call from an unknown number from a man in police uniform who said he is Inspector Vikram Rathore from CB, Mumbai. He said the complainant's Aadhaar card was used to open a bank account for money laundering and an arrest warrant is issued. He was not allowed to disconnect the call and was told not to tell anyone in the family. He transferred Rs. 35,46,500 through RGS to account #0230517360305.

FIR no: 80275026N
Section: BNS 396, 398D, 3194G, 3192D IT Act 68D
Complainant: Kamlesh B. Mahabadi, age 57, textile trader, c/o Satellite Road, Mha. 0000000206

Registered: 24 Sep 2026
Place: Prantahadgaon, about 1.5 km west of P.S.

Evidence identifiers

DATA ACCESS: 40230517360305

Processing

CrimeFIR

All investigations & crime pattern detector

Dashboard

Report FIRs

Case Files

Report Offenders

Investigation Graphs

Station Briefs

Model Quality

Reported near-offender cluster

K-001

7 FIRs across 6 police stations in 3 districts; 24 Jul 2026 - 24 Sept 2026, total reported loss Rs 2,30,32,000.

Search: FIRs, phone, bank account, suspects

ALL FIRs

Case: CTS

Dist: Satara - CTS

All Crime Categories

Strong: Time Only

Investigation Graph

Graph showing connections between FIRs and evidence. Nodes include: SAR-804-2026-0241, AMP-SAT-2026-0236, AMP-CYS-2026-0258, SAR-804-2026-0241, AMP-SAT-2026-0236, VAD-SAY-2026-0295, AMP-SAT-2026-0236.

Legend: SAFETY (Green) & INDICATORS (Red)
FIR Case (Blue), Bank Account (Orange), Victim Profile (Green), Suspect Name (Red)
Strong (Green) & Time (Red), Potential (Yellow)

Timeline of linked FIRs

Key Identifiers

Bank account - in 4 FIRs - centrality 3.084
40230517360305
Bank account - in 3 FIRs - centrality 0.223
410000000100001
Phone - in 1 FIR - centrality 0.222
0577040100000

Identifies the offender (signature only)
Vikram Rathore (3 FIRs)

Risk factors

7 FIRs linked by shared evidence (x-10)
opens 8 police stations (x-10)
creates 3 districts (x-10)
total loss Rs 23,032,000 (x-10)
action in the last 30 days of data (x-10)
1 victim (victim identity) (x-10)
money trail shared bank account / UPI ID (x-10)

Suggested next actions

Request Police and KYC/Transaction lead for bank account #0230517360305 (Bank / KYC)
Request Police and KYC/Transaction lead for bank account #0230517360305 (Bank / KYC)
Request CB, subscriber details and IMEI history for +877062419038 (Informant provided)
Share this cluster with the 6 involved stations for a joint investigation
Investigation lead only. Requires human verification.

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Station brief and measured quality



Measured, not claimed

Output	Result
Crime major / minor head	93.4% / 83.0% (macro-F1 0.816)
Laya (accepted at >= 0.40) / Granite (escalated)	80.4% on 265 FIRs / 40 of 40 correct
Hard evidence extraction	100% precision, 100% recall, 100% roles
Repeat-offender clusters	100% precision, 93.7% recall, 11/12 gangs, 0 decoys
Accused names	88% found
MO flags (weakest output)	F1 0.64, 6 unreliable flags disabled
Team's first prototype, on its own 350-FIR file	found 1 of 12 repeated phones; CrimeFIR finds 12/12
Throughput, RTX 3050 laptop 4 GB	400 FIRs ~6 min; live batch of 6 in 14 s

IBM technology is load-bearing

IBM Bob

MCP server: 12 tools (search FIRs, related FIRs, flagged clusters, station trends & briefs)
'FIR Analyst' custom mode + rules
bob run: headless weekly brief

IBM Granite on watsonx.ai

granite-4-h-small (eu-de)
System-2 second opinion
grounded JSON extraction
station brief prose, numbers
verified against the data

IBM Granite Embedding

granite-embedding-30m-english
local GPU, CPU fallback
narrative vectors for
pattern-only leads

- Investigator asks Bob: "Which high-risk clusters touched Navrangpura this month, and what first?"
- Bob calls CrimeFIR tools and answers with FIR / cluster ids, never asserting guilt.
- Bob Shell runs the weekly station brief headlessly: bob run --format json --max-cost 2.

Impact and next steps

- Plugs onto existing FIR text: no change to how FIRs are written; outputs map to NCRB forms.
- Surfaces inter-district rings in minutes: faster freeze requests on mule accounts, joint investigations.
- Scale path: PostgreSQL + pgvector, model-service replicas, batched Laya on a GPU server.
- Next: authentication and roles, Indian-language FIRs (Laya multilingual), officer feedback to retrain thresholds.

Honest limitations

synthetic (real-case-grounded) data · names without surname are not
provably one person · MO flags weakest · pattern links weak · no auth yet · all links are leads needing verification